

Power BI Project Final Report

Business Information Systems (BIS)

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Abstract

This project focuses on analyzing a technology advertising campaigns dataset using Power BI. The main goal was to extract meaningful insights that can help businesses optimize their marketing strategies and improve performance. We performed data cleaning, transformation, and modeling, followed by creating interactive dashboards. The analysis revealed key trends related to campaign performance, cost efficiency, and conversion rates, which can support better decision-making.

Contents

1	Introduction	4
2	Dataset Description	4
2.1	Dataset Features	4
2.2	Dataset Size	4
2.3	Business Problem	4
3	Data Cleaning (Power Query / ETL)	4
3.1	Cleaning Steps	5
3.2	Data Transformation	5
4	Data Modeling	5
4.1	Model Components	5
4.2	Date Table	5
5	DAX Measures	5
5.1	Key Measures	5
5.2	Calculated Metrics	6
6	Dashboard Development	6
6.1	Dashboard Components	6
6.2	User Interaction	6
7	Business Insights	6
7.1	Key Findings	6
7.2	Detailed Analysis	6
8	Challenges Faced	7
9	Conclusion	7
10	Recommendations	7

1 Introduction

In today's data-driven world, businesses rely heavily on analytics to make strategic decisions. This project aims to utilize Power BI to analyze advertising campaign data and transform raw data into valuable insights.

The dataset represents multiple advertising campaigns across different platforms, including information such as impressions, clicks, cost, conversions, and revenue. By analyzing this data, we aim to understand performance patterns and identify areas for improvement.

2 Dataset Description

The dataset used in this project is a technology advertising campaigns dataset. It contains information about multiple campaigns conducted across different platforms.

2.1 Dataset Features

The dataset includes the following important columns:

- Campaign ID
- Date
- Platform (Google, Facebook, etc.)
- Impressions
- Clicks
- Cost
- Conversions
- Revenue

2.2 Dataset Size

The dataset contains hundreds to thousands of rows representing daily campaign performance, making it suitable for analysis and visualization.

2.3 Business Problem

The main problem addressed in this project is:

How can companies optimize their advertising campaigns to maximize revenue while minimizing cost?

3 Data Cleaning (Power Query / ETL)

Before building the dashboard, we performed several data cleaning steps using Power Query.

3.1 Cleaning Steps

- Removed null and missing values.
- Removed duplicate rows.
- Converted data types (e.g., Date to Date format, Cost to Decimal).
- Renamed columns for better readability.
- Filtered irrelevant or incorrect data.

3.2 Data Transformation

- Created calculated columns such as:
 - Click Through Rate (CTR)
 - Conversion Rate
- Formatted date columns for time-based analysis.

4 Data Modeling

We built a relational data model in Power BI using a star schema approach.

4.1 Model Components

- Fact Table: Campaign Performance Data
- Dimension Table: Date Table

4.2 Date Table

We created a date table using DAX:

```
Calendar = CALENDARAUTO()
```

5 DAX Measures

To analyze the dataset, we created several DAX measures.

5.1 Key Measures

```
Total Revenue = SUM(Revenue)
```

```
Total Cost = SUM(Cost)
```

```
Total Clicks = SUM(Clicks)
```

```
Total Conversions = SUM(Conversions)
```

5.2 Calculated Metrics

$\text{CTR} = \text{DIVIDE}(\text{SUM}(\text{Clicks}), \text{SUM}(\text{Impressions}))$

$\text{Conversion Rate} = \text{DIVIDE}(\text{SUM}(\text{Conversions}), \text{SUM}(\text{Clicks}))$

$\text{ROI} = \text{DIVIDE}(\text{SUM}(\text{Revenue}) - \text{SUM}(\text{Cost}), \text{SUM}(\text{Cost}))$

6 Dashboard Development

We designed an interactive dashboard in Power BI to visualize the data.

6.1 Dashboard Components

- KPI Cards (Revenue, Cost, ROI)
- Line Charts (Performance over time)
- Bar Charts (Platform comparison)
- Pie Charts (Revenue distribution)

6.2 User Interaction

The dashboard includes filters for:

- Date
- Platform
- Campaign ID

7 Business Insights

From the dashboard, we extracted several key insights.

7.1 Key Findings

- Campaign performance varies significantly across platforms.
- Some campaigns generate high impressions but low conversions.
- ROI is highest when cost is optimized rather than increased.
- Certain time periods show peak performance.

7.2 Detailed Analysis

We observed that campaigns on specific platforms (e.g., Google Ads) tend to have higher conversion rates compared to others. Additionally, campaigns with better targeting result in higher ROI.

8 Challenges Faced

During the project, we encountered several challenges:

- Handling missing data
- Understanding DAX formulas
- Building relationships between tables
- Designing meaningful dashboards

9 Conclusion

This project demonstrated the power of data analysis using Power BI. By transforming raw data into meaningful insights, businesses can make informed decisions and improve their marketing strategies.

10 Recommendations

Based on our analysis, we recommend:

- Focus on high-performing platforms.
- Optimize cost allocation across campaigns.
- Improve targeting strategies.
- Continuously monitor performance using dashboards.